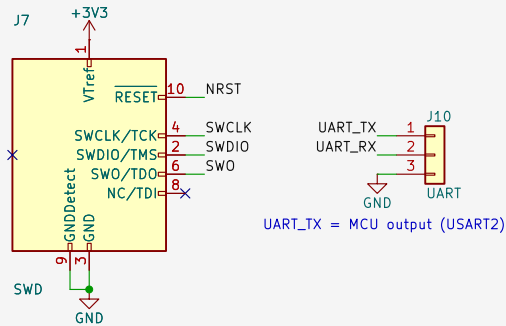
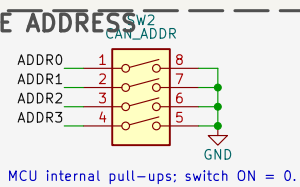


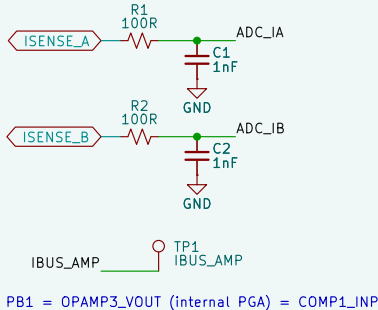
## DEBUG: SWD + UART



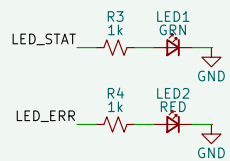
## CAN NODE ADDRESS



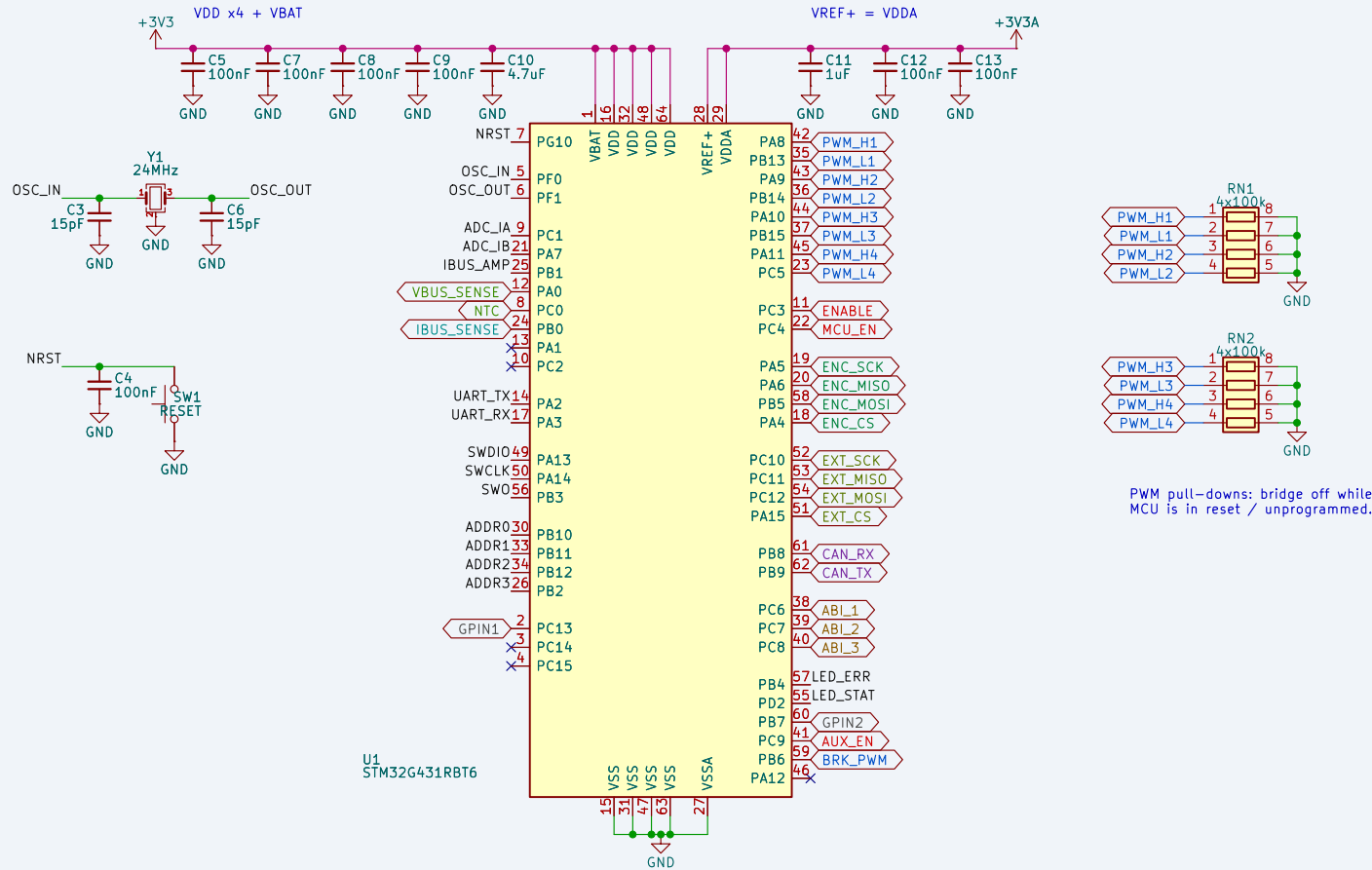
## ADC INPUT RC



## STATUS LEDS



## MCU STM32G431RBT6



## DESIGN NOTES (Rev B)

- Option bytes: nSWBOOT0 = 0 (PB8 = BOOT0 is shared with FDCAN1\_RX).
- TIM1 complementary PWM, CH1-CH4 + CH1N-CH4N, centre aligned 30 kHz (20-50 kHz). DTG 50-100 ns is a backup only; DRV8300D inserts a fixed 200 ns dead time.
- Duty limit 97-98 % (bootstrap refresh). Pre-charge bootstraps (low sides on) after every enable and after Hi-Z.
- Hardware trips: RS3 -> PB0 OPAMP3 (PGA) -> PB1 -> COMP1 -> TIM1\_BKIN. IA PC1 -> COMP3, IB PA7 -> COMP2 (DAC thresholds). ENABLE PC3 -> TIM1\_BKIN2.
- ADC: IA PC1 = ADC1\_IN7 and IB PA7 = ADC2\_IN4 (simultaneous). VBUS PA0, NTC PC0.
- No brake chopper fitted: limit regen via VBUS monitoring (Q9/J9 footprint is DNP).
- External encoder uses SPI3 (SPI2 SCK only on PB13/PF1, both in use).
- ABI / Hall on TIM3 CH1-CH3 (encoder mode or Hall XOR interface).

### Power Supply

File: power\_supply.kicad\_sch

### Power Stage

File: power\_stage.kicad\_sch

### Interfaces

File: interfaces.kicad\_sch

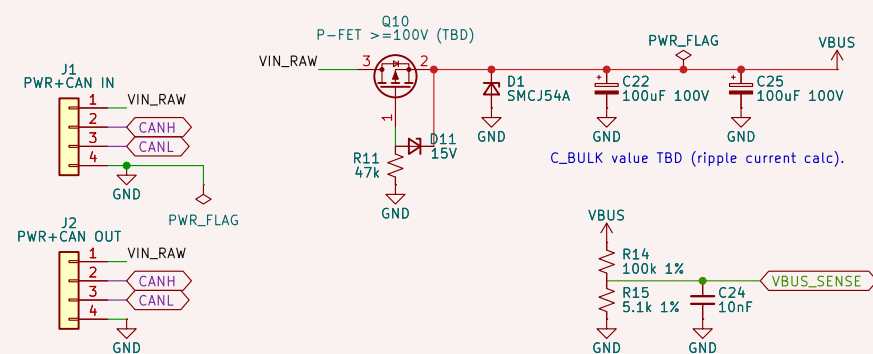
### FOC Motor Driver

Sheet: /  
File: foc-driver.kicad\_sch

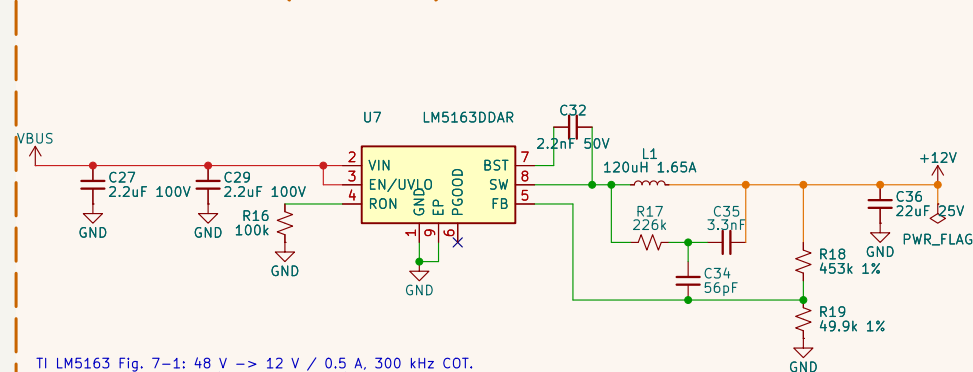
### Title: FOC Motor Driver Rev B - MCU

Size: A3 Date: 2026-10-02  
KiCad E.D.A. 10.0.6

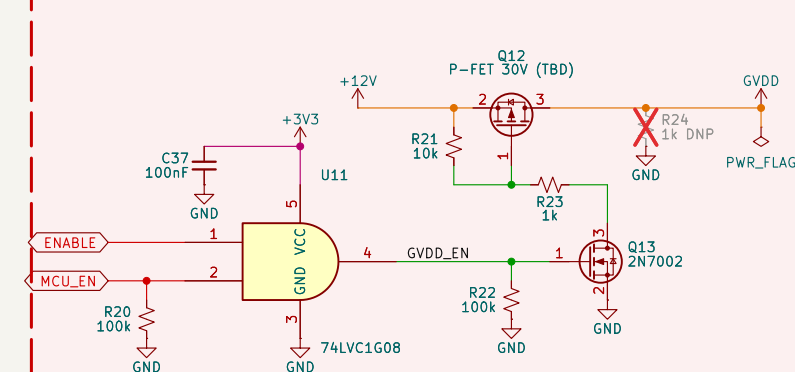
Rev: B  
Id: 1/4



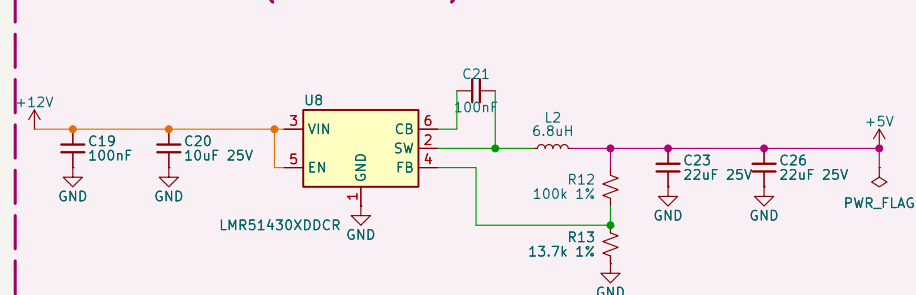
Daisy chain: J1/J2 in parallel.  
Current rating = whole chain.  
Pinout proposal (connector TBD).



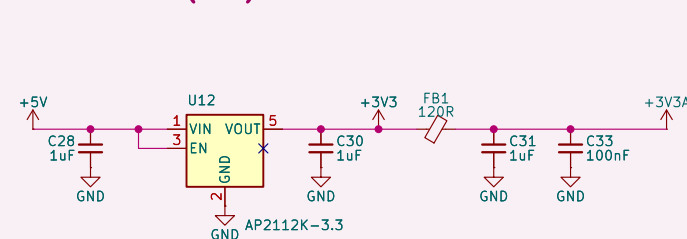
TI LM5163 Fig. 7-1: 48 V → 12 V / 0.5 A, 300 kHz COT.  
VIN/EN abs max 100 V (TVS clamp ~87 V).



ENABLE low OR MCU\_EN low -> GVDD off -> DRV8300 UVLO -> all FETs off  
(independent of MCU; TIM1\_BKIN2 is the fast path).  
GVDD decay time set by driver caps; R DNP = optional bleeder.

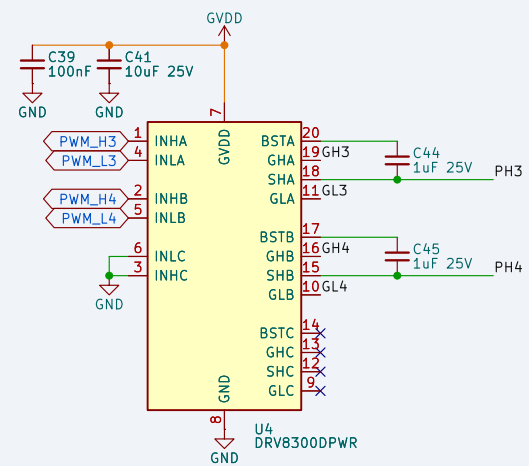
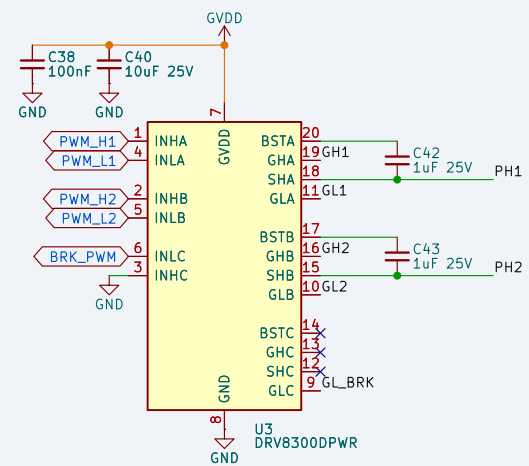


500 kHz variant, datasheet Table 9–1 (5 V: 6.8  $\mu$ H, 2x22  $\mu$ F, 100k/13.7k).  
5 V feeds CAN transceiver VCC and J8 encoder supply.



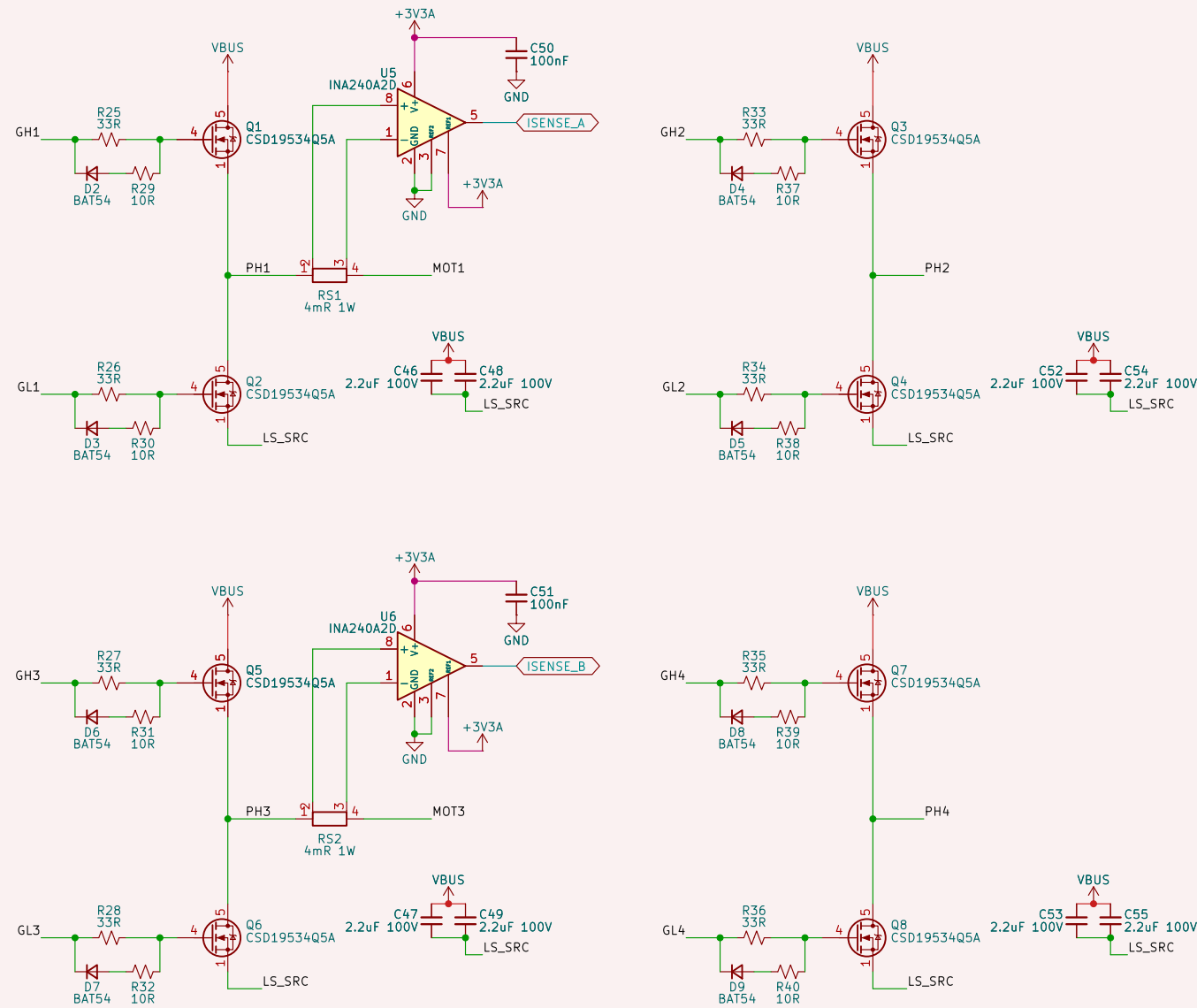
+3V3: MCU, CAN VIO, logic (~80 mA).  
+3V3A: VDDA/VREF+, INA240, AS5047D, NTC.

## GATE DRIVERS

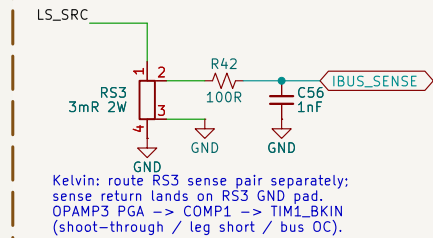


DRV8300D TSSOP: fixed 200 ns dead time + cross-conduction lockout.  
MCU TIM1 DTG is a backup (50–100 ns). CBST <= 1 uF.  
U3 channel C: low side only -> optional brake chopper Q9 (DNP).

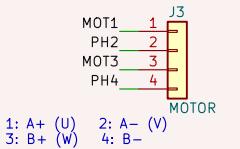
## HALF BRIDGES + IN-LINE CURRENT SENSE



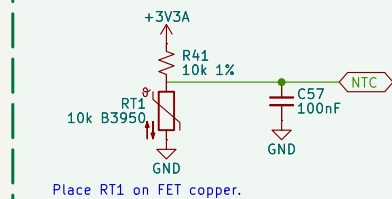
## BUS SHUNT RS3



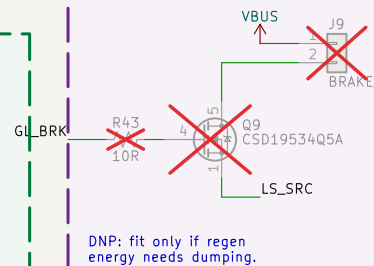
## MOTOR



## FET TEMP



## BRAKE CHOPPER (DNP)



## FOC Motor Driver

Sheet: /Power Stage/  
File: power\_stage.kicad\_sch

## Title: Power Stage

Size: A3 Date: 2026-10-02  
KiCad E.D.A. 10.0.6

Rev: B  
Id: 3/4

